

Final Project Assignment: Cleaning robot on a tilting platform

In this project, your assignment is to design and control a manipulator for cleaning purposes.

The manipulator is a 6-DoF Unitree Z1 robot mounted on a pan-tilt platform designed to increase its workspace.

The task is to track a circular horizontal trajectory of 20 cm radius around a position (which involves motion of the end-effector out of the manipulator workspace), while keeping the end-effector perpendicular to a table.

If the platform is actuated, the robot becomes redundant (8-DoF) for the task and it can:

- 1) perform the task
- 2) exploit the redundancy (2 DoF) to achieve a secondary task, like maximizing the dexterity (i.e. manipulability [1]) of the robot's end-effector.

The project involves the following steps:

1. create a URDF model of the assembly manipulator + platform (roll/pitch joints)
2. setup a simulator for the robot inside the Locosim framework (to compute the kinematics you can use the Pinocchio library)
3. Generate the 20cm radius horizontal reference trajectory for the end-effector (with orientation perpendicular to the table)
4. Design a task space inverse-dynamics to track the reference trajectory (primary task)
5. Implement a secondary task in the null-space of the primary task to maximize the dexterity.

[1] "Robotics Modelling, Planning and Control" - Siciliano, B., Sciavicco, L., Villani, L., Oriolo, G.

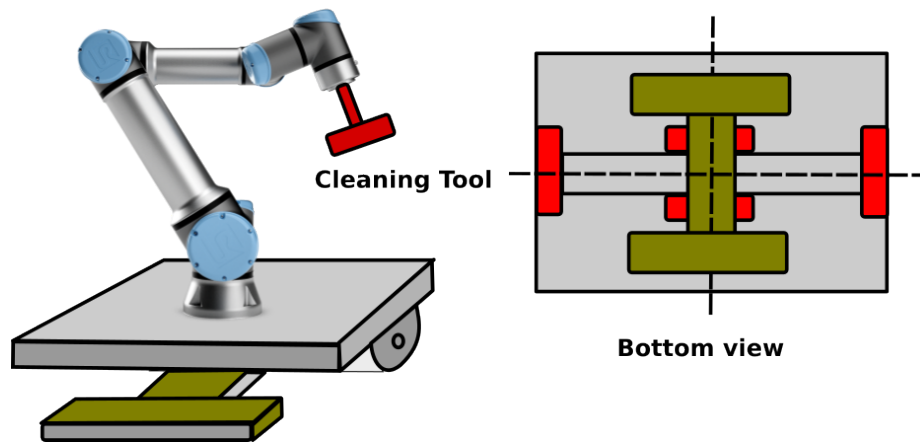


Figure 1: drawing